



ENT-164 · Intro to Making

Tufts
UNIVERSITY

Derby Entrepreneurship
Center at Tufts

CLASS 9 · WEEK 9 · FALL 2026

Final Project Kickoff

The Robo Challenge is in the books. Today we celebrate what you built, then launch the project you'll present on Demo Day — brainstorm ideas, pitch a concept, and get feedback from your peers.

Thu, Nov 5

JCC 301 / Nolop

Concept generation

Note-and-Vote



Robo Challenge awards

You designed, built, and raced working robots — here's how it ended.

The race

Teams moved as many balls as possible between the two lines, with robots they built from scratch.

The results

Scores are in, and the awards go out to the teams behind this semester's robots.

Carry it forward

Reflect on what worked, what didn't, and what you'd do differently — then bring that into the final project.

team	score
orange	$(19 \times 2) + (19 \times 2) + (18 \times 2) + (16 \times 2) + (18 \times 2) + (20 \times 2) + (17 \times 2) = 254$
pink	$(17 + 15 + 15) \times 2 = 94$
green	$11 + 8 = 19$
teal	$(17 \times 5) = 85$

DUE TOMORROW

Mid-term **self & peer reviews**

Last call — no later than tomorrow at noon, Friday, Nov 6.

01

Midterm self review

Reflect on your own contributions to the Robo Challenge: what you built, how you worked with your team, and what you would do differently.

02

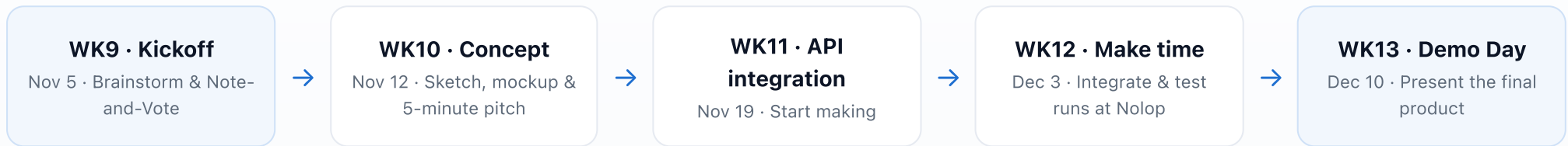
Midterm peer reviews

Give every teammate honest, specific feedback on the first half of the semester.

DEADLINE Tomorrow, Friday, Nov 6 at noon. Links and instructions are on Canvas — don't leave it to the last minute.

Five weeks to Demo Day

What happens between today and the final showcase.

**DELIVERABLE**

Concept presentation — 5 minutes per team: problem & opportunity, proposed product concept with sketch and mockup, and next steps.

No class Thu, Nov 26 (Thanksgiving). Build sessions happen at Nolop.

SAVE THE DATE

Demo Day: **show what you built**

Final presentations and live product demos, built by students with no prior experience. Open to the public — no registration required.

Thu, Dec 10

11:30 am

JCC 160

Week 13 · the finish line for your final project.

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ENT164 Introduction to Making

Demo Day!

See the final presentations & live product demos built by students with no prior experience. Open to the public.



What your build **must include**

Your concept proposal has to show how you'll hit every one of these.

01

Laser cutting

At least one part designed as a 2D cut — and the proposal says where it fits in the build.

02

3D printing

Printed parts where they make sense: mounts, brackets, and custom shapes.

03

Microcontroller + I/O

A microcontroller with sensors and actuators, so the product senses the world and responds.

04

Smart API integration

AI or another external service wired into the device — a core maker skill.

THE BRIEF

Solves a unique problem — and fits in the project box: **40 × 25 × 25 cm**. Your proposal should clearly indicate which parts use each method and how they contribute to the design.

** Cannot be an automatic plant-watering system :)

The 5-minute **concept pitch**

Every team presents its final project concept to the class.

01

Problem & opportunity

A clear articulation of the problem and why it matters.

02

Proposed product concept

A walkthrough of the concept and how it meets the final product requirements — bring sketches and a foam-core or other mockup.

03

Next steps

What you will build and test in the second half of the semester.

WORKING DOCS

Team folders for working docs and presentations: go.tufts.edu/teams

Note-and-Vote: silent ideas, fast decision

01 • NOTE

Ideas first, alone

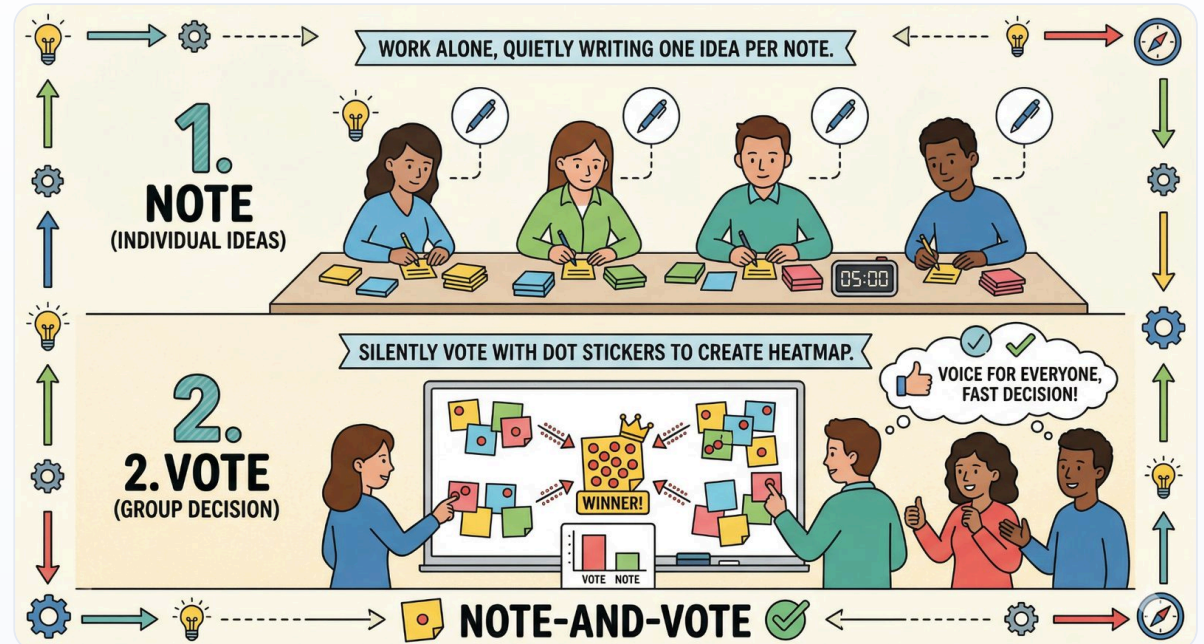
Everyone writes their own ideas quietly, one per note, so the loudest voice doesn't set the agenda.

02 • VOTE

Vote with dots

Post the notes, then vote in silence with round stickers. The heatmap shows where the team's energy really is.

Voice for everyone, fast decision — then the team discusses the top ideas together.



From sticky notes to a team concept

STEP 1

Individually · 10 min · in silence

Write ideas

- Write 3 unique ideas
- One idea per sticky note
- Each idea must meet the criteria

STEP 2

As a team · 5 min · in silence

Post & vote

- Post every idea on the team wall
- Vote for your top 3 with round stickers
— you can vote for your own

STEP 3

As a team · 20 min

Discuss & decide

- Count the votes
- Discuss the top 3 ideas
- Decide what you'll pursue

THEN Go directly to **Nolop** and start the concept presentation — sketch, mockup, and slides.

New project **boxes**

Your team's final project starts with a fresh box of parts.

Everything in one box

Pick up your team's project box — the parts for the final build live here.

It has to fit back in

The finished product must fit inside the box: **40 × 25 × 25 cm**. Plan for the size from day one.



BEFORE YOU GO

Questions?

Next class: concept presentations, Thu, Nov 12.

WHERE TO FIND US

Office hours

go.tufts.edu/meet_tvdv

Email

thomas.van_de_velde@tufts.edu

Team folders

go.tufts.edu/teams

Course materials

tuftsmaker.github.io/ENT-164